

100% Excellent!

PHY201 Exam on chapter(s): 1, 2, 3

Name: 9

These questions assess your understanding of the material from Chapters 1, 2, 3 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. To test the idea of surviving a free-falling elevator, Joshua conducts an experiment using a string-attached plank and a programmable toy flea. His makeshift elevator falls 24.2 m and the flea jumps 0.0003 s before impact. Previous tests show that these toys can endure impacts below 6.41 m/s. What is the minimum initial speed needed for the flea's vertical jump to make it impact the ground at a safe speed? (Note: The plank's mass is significantly greater than the toy's mass.)

( $\vec{v}_{E,G_i}$ )  
 $\vec{v}_E$  just before impact

$$\vec{v}_{yf}^2 = \vec{v}_{yi}^2 + 2\vec{a}_y\Delta y$$

$$\vec{v}_{yf} = \sqrt{2\vec{a}_y\Delta y}$$

$$\vec{v}_{yf} = \sqrt{2(-9.8 \frac{m}{s^2})(-24.2 m)}$$

$$\vec{v}_{yf} = 21.779 \frac{m}{s}$$

$$\vec{v}_{yf} = \vec{v}_{yi} + \vec{a}_y t$$

$$t = \frac{\vec{v}_{yf}}{\vec{a}_y} = \frac{-21.779 \frac{m}{s}}{-9.8 \frac{m}{s^2}} = 2.222 s$$

$\vec{v}_f$  when frog jumps

$$t_i = t_f - 0.0003 s$$

$$\vec{v}_{yf} = \vec{v}_{yi} + \vec{a}_y t_i$$

$$\vec{v}_{yf} = (-9.8 \frac{m}{s^2})(2.222 s - 0.0003 s)$$

$$\vec{v}_{yf} = -21.776 \frac{m}{s}$$

$$\vec{v}_{f,G_i} = \vec{v}_{f,G_i} + \vec{a}_y t$$

$$\vec{v}_{f,G_i} < -6.41 \frac{m}{s}$$

$$\vec{v}_{f,G_i} + \vec{a}_y t < -6.41 \frac{m}{s}$$

ANSWER: 15.37  $\frac{m}{s}$ 

$$\vec{v}_{E,f} = \vec{v}_{E_i} + \vec{a}_y t_i = -21.776 \frac{m}{s} \quad a_{f,E} = a_{f,G} - a_{E,G}$$

$$\vec{v}_{f,G_i} + -21.776 \frac{m}{s} < -6.41 \frac{m}{s} \quad a_{f,G} = a_{f,E} + a_{E,G}$$

$$-\vec{v}_{f,G_i} + 21.776 \frac{m}{s} > 6.41 \frac{m}{s}$$

$$\text{speed} = |-\vec{v}_{f,G_i}| > 15.366 \frac{m}{s}$$

$$\text{speed} > \boxed{15.366 \frac{m}{s}}$$

2. Stephen kicks a soccer ball, initially at rest on the ground, with an initial velocity of 10.5 m/s at a launch angle of 45.2° above the horizontal. The ball flies through the air and lands back on the ground. Calculate its total flight time, assuming that air resistance is negligible.

10.5 m/s  
 45.2°  
 $\Delta x = R = \frac{v_i^2 \sin(2\theta_i)}{g}$

$$\vec{v}_{x_i} = \vec{v}_i \cos(\theta_i)$$

$$\vec{x}_f = \vec{x}_i + \vec{v}_{x_i} t + \frac{1}{2} \vec{a}_x t^2$$

$$\Delta x = \vec{v}_{x_i} t$$

$$t = \frac{\Delta x}{\vec{v}_{x_i}} = \frac{v_i^2 \sin(2\theta_i)}{g}$$

$$(10.5 \frac{m}{s})^2 \sin(2 \cdot 45.2^\circ)$$

$$t = \frac{9.8 \frac{m}{s^2}}{(10.5 \frac{m}{s}) \cos(45.2^\circ)} = \boxed{1.52 s}$$

ANSWER: 1.52 s

3. On a family trip, John and Abigail take their son, Ryan, to visit the deepest man-made hole on Earth, the Kola Superdeep Borehole. Ryan shows little interest, however, and spends all his time on his cell phone. John, fairly horrified at his son's lack of scientific interest, grabs the phone and throws it down the hole, which is 9 inches in diameter and 12,262 meters deep. John manages to hear over Ryan's cries that it hit the bottom in 51.1 seconds. What was the velocity at which John threw the cell phone?

$$\vec{y}_f = \vec{y}_i + \vec{v}_{y_i} t + \frac{1}{2} \vec{a}_y t^2$$

$$\vec{v}_{y_i} = \frac{-\vec{y}_f - \frac{1}{2} \vec{a}_y t^2}{t}$$

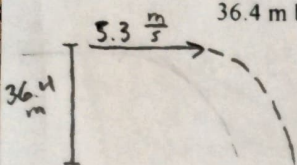
$$\vec{v}_{y_i} = \frac{-(12262 m) - \frac{1}{2} (-9.8 \frac{m}{s^2}) (51.1 s)^2}{51.1 s} = \boxed{10.43 \frac{m}{s}}$$

ANSWER: 10.43  $\frac{m}{s}$ 

+3



4. A great blue heron is flying horizontally at a speed of 5.3 m/s when the perch in her talons wiggles loose and falls into the lake 36.4 m below. Calculate the speed at which the fish hits the water.



ANSWER:  $27.23 \frac{m}{s}$

$$speed = |\vec{v}_f| = \sqrt{\vec{v}_{xf}^2 + \vec{v}_{yf}^2}$$

$$|\vec{v}_f| = \sqrt{(5.3 \frac{m}{s})^2 + (26.71 \frac{m}{s})^2} = \boxed{27.23 \frac{m}{s}}$$

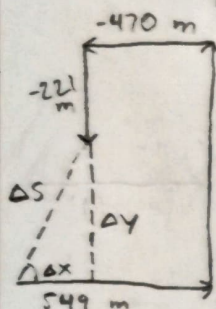
$$\vec{v}_{xi} = \vec{v}_{xf}$$

$$\vec{v}_{yf}^2 = \vec{v}_{yi}^2 + 2\vec{a}_y\Delta y$$

$$\vec{v}_{yf} = \sqrt{2(-9.8 \frac{m}{s^2})(-36.4 m)}$$

$$\vec{v}_{yf} = 26.71 \frac{m}{s}$$

5. It takes 125 seconds for a horse to move 549 m east, then 857 m north, then 470 m west, and finally 221 m south. What is the horse's net displacement? Include both magnitude and direction.



$$\Delta x = 549 m - 470 m = 79 m$$

$$\Delta y = 857 m - 221 m = 636 m$$

$$\Delta s = \sqrt{\Delta x^2 + \Delta y^2} = \sqrt{(79 m)^2 + (636 m)^2} = \boxed{640.89 m}$$

$$\tan \theta = \frac{|\Delta y|}{|\Delta x|}$$

$$\theta = \tan^{-1}\left(\frac{636 m}{79 m}\right) = \boxed{82.92^\circ}$$

ANSWER:  $640.89 m, 82.92^\circ N \text{ of } E$

6. A basketball referee tosses the ball straight up for the starting tip-off. What is the minimum velocity at which a basketball player must leave the ground in order to rise 2.7 m above the floor and make contact with the ball?

$$\vec{v}_{yf}^2 = \vec{v}_{yi}^2 + 2\vec{a}_y\Delta y$$

$$\vec{v}_{yi} = \sqrt{-2\vec{a}_y\Delta y}$$

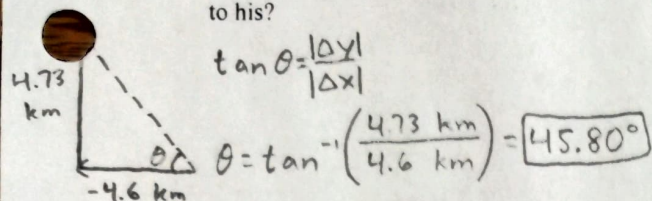
$$\vec{v}_{yi} = \sqrt{-2(-9.8 \frac{m}{s^2})(2.7 m)} = \boxed{7.27 m/s}$$

ANSWER:  $7.27 \frac{m}{s}$



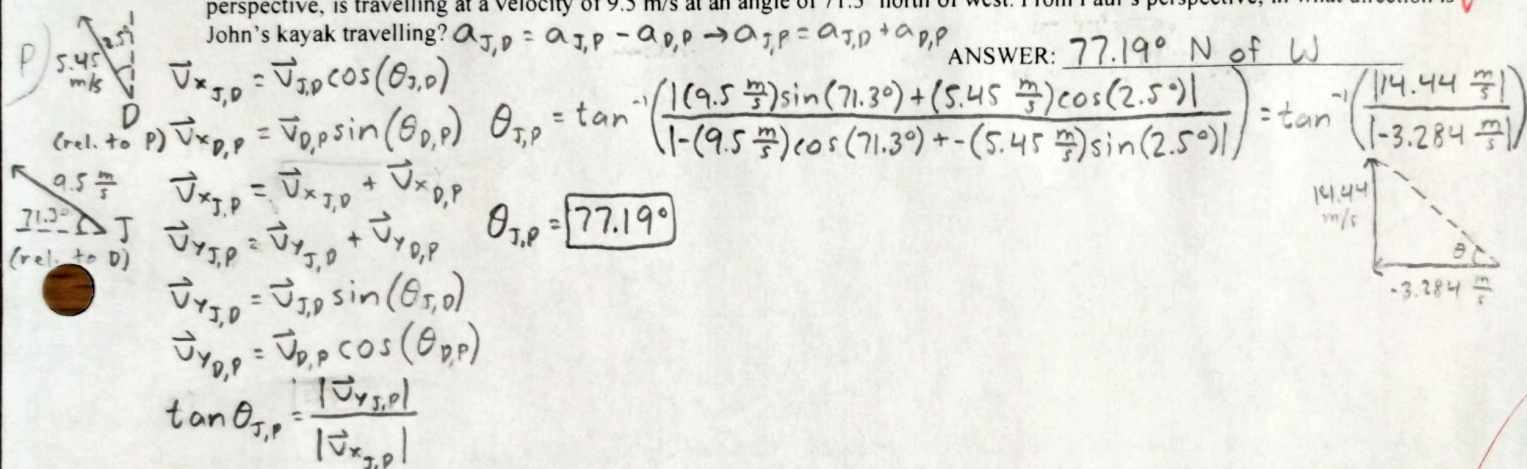
7. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Stephen's. To get to Stephen's, you have to walk 4.6 km west and then 4.73 km north. What is the compass direction of a line connecting your house to his?

ANSWER: 45.80° N of W



8. Paul is sitting in his folding chair on the beach, watching his friend Dylan as he rides in a dinghy moving at a velocity of 5.45 m/s at an angle of 2.5° west of north. Dylan then passes his brother, John, who is riding in a kayak, which, from Dylan's perspective, is travelling at a velocity of 9.5 m/s at an angle of 71.3° north of west. From Paul's perspective, in what direction is John's kayak travelling?  $\vec{a}_{J,P} = \vec{a}_{J,D} - \vec{a}_{D,P} \rightarrow \vec{a}_{J,P} = \vec{a}_{J,D} + \vec{a}_{D,P}$

ANSWER: 77.19° N of W



9. A small airplane flies in a straight line at a speed of 209 km/h. How long does it take the plane to fly 113 leagues? Report your answer in minutes. An inch is 2.54 cm, a mile is 2640 cubits, a league is 3 miles, and a cubit is 2 feet.

ANSWER: 156.62 min.

$$\frac{113 \text{ leagues}}{1 \text{ league}} \times \frac{3 \text{ mi}}{1 \text{ mi}} \times \frac{5280 \text{ ft}}{1 \text{ ft}} \times \frac{12 \text{ in.}}{1 \text{ in.}} \times \frac{2.54 \text{ cm}}{1 \text{ cm}} \times \frac{1 \text{ km}}{10^5 \text{ cm}} = 545.568 \text{ km}$$

$$\vec{v} = \frac{\Delta \vec{x}}{t}$$

$$t = \frac{\Delta \vec{x}}{\vec{v}} = \frac{545.568 \text{ km}}{209 \frac{\text{km}}{\text{h}}} = \frac{545.568 \text{ km}}{3.483 \frac{\text{km}}{\text{min.}}} = 156.62 \text{ min.}$$

$$\frac{209 \text{ km}}{1 \text{ h}} \times \frac{1 \text{ h}}{60 \text{ min.}} = 3.483 \frac{\text{km}}{\text{min.}}$$



10. Shiv throws a 3.9-g coin straight downward from a 24-story window with a vertical speed of 2.8 m/s (approximately 6.3 mph). Ignoring the effect of the drag force, at what speed does the coin impact a patch of grass on the ground? Express the speed in meters per second. Take a story as roughly 10.3 feet. There are 3.28 feet in one meter.

$$\frac{24 \text{ stories}}{1 \text{ story}} \times \frac{10.3 \text{ ft}}{3.28 \text{ ft}} = 75.366 \text{ m}$$

ANSWER:  $38.54 \frac{\text{m}}{\text{s}}$

75.366 m

$$\vec{v}_y^2 = \vec{v}_{y_i}^2 + 2\vec{a}_y \Delta y$$

$$\vec{v}_y = \sqrt{(-2.8 \frac{\text{m}}{\text{s}})^2 + 2(-9.8 \frac{\text{m}}{\text{s}^2})(-75.366 \text{ m})} = 38.54 \frac{\text{m}}{\text{s}}$$

11. Tyler, a fireman by profession but physicist at heart, jumps horizontally from a burning building with an initial speed of 7.9 m/s. Later, Tyler is interested in determining the time at which the angle between his velocity and acceleration vectors is  $49^\circ$ . At what time does that happen?

ANSWER:  $0.7008 \text{ s}$

7.9  $\frac{\text{m}}{\text{s}}$

$$\tan \theta = \frac{|\vec{v}_{x_f}|}{|\vec{v}_{y_f}|} = \frac{v_{x_f}}{a_y t}$$

$$t = \frac{v_{x_f}}{\tan \theta} = \frac{7.9 \frac{\text{m}}{\text{s}}}{\tan(49^\circ)} = 0.7008 \text{ s}$$

$\vec{v}_{x_f} = \vec{v}_{x_i}$   
 $\vec{v}_{y_f} = \vec{v}_{y_i} + \vec{a}_y t$

12. Sam throws a book horizontally from his dorm window 26.3 m above the ground. Caleb, another student, stands 6.42 m away from the dorm building and catches the book at a height of 1.72 m above the ground. Caleb claims that Sam threw the book too slowly, so Sam wants to calculate its initial speed to prove him wrong. Ignoring friction, what initial speed does he calculate?

ANSWER:  $2.87 \text{ m/s}$

26.3 m

6.42 m

1.72 m

$$\vec{y}_f = \vec{y}_i + \vec{v}_{y_i} t + \frac{1}{2} \vec{a}_y t^2$$

$$t = \sqrt{\frac{2(\vec{y}_f - \vec{y}_i)}{\vec{a}_y}}$$

$$\vec{x}_f = \vec{x}_i + \vec{v}_{x_i} t + \frac{1}{2} \vec{a}_x t^2$$

$$\vec{v}_{x_i} = \frac{\vec{x}_f}{t} = \frac{6.42 \text{ m}}{\sqrt{\frac{2(1.72 \text{ m} - 26.3 \text{ m})}{-9.8 \frac{\text{m}}{\text{s}^2}}}} = 2.87 \frac{\text{m}}{\text{s}}$$